

Code No: 182AR**JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD****B. Tech I Year II Semester Examinations, January/February - 2024****ORDINARY DIFFERENTIAL EQUATIONS AND VECTOR CALCULUS****(Common to CE, EEE, ME, ECE, CSE, EIE, IT, AE, MIE, CSIT, CE(SE), CSE(CS), CSE(AI&ML), CSE(DS), CSE(IOT), AI&DS, AI&ML, CSD)****Time: 3 Hours****Max. Marks: 60****Note:** This question paper contains two parts A and B.i) **Part- A** for 10 marks, ii) **Part - B** for 50 marks.

- Part-A is a compulsory question which consists of ten sub-questions from all units carrying equal marks.
- Part-B consists of **ten questions** (numbered from 2 to 11) **carrying 10 marks each**. From each unit, there are two questions and the student should answer one of them. Hence, the student should answer five questions from Part-B.

PART- A**(10 Marks)**

- Write down the general form of a linear differential equation of first order. [1]
- Give an example of a family of curves which is self-orthogonal. [1]
- Define the Legendre's homogeneous differential equation. [1]
- Find the particular integral of $\frac{d^3y}{dx^3} + 4\frac{dy}{dx} = \sin 2x$. [1]
- Show that the Laplace transform of the unit step function $u(t - a)$ is $\frac{e^{-as}}{s}$. [1]
- State the Convolution Theorem for Laplace transform. [1]
- Define a vector point function give an example. [1]
- Find the curl of $\vec{V} = e^{xyz}(\vec{i} + \vec{j} + \vec{k})$ at the point $(1, 2, 3)$. [1]
- State Stoke's Theorem. [1]
- If S is a closed surface and V is the volume of the region bounded by S then what is the value of $\oint_S \vec{r} \cdot \vec{N} dS$ where $\vec{r} = x\vec{i} + y\vec{j} + z\vec{k}$. [1]

PART - B**(50 Marks)**

- A bacterial population B is known to have a rate of growth directly proportional to B itself. If between noon and 2 PM the population triples, at what time, no controls being exerted, should B become 100 times what it was at noon.
- Solve $x \left[\frac{dx}{dy} + y \right] = 1 - y$. [5+5]

OR

- Solve $[1 + \log(xy)]dx + \left[1 + \frac{x}{y}\right]dy = 0$.
- Find the particular member of the orthogonal trajectories of $x^2 + cy^2 = 1$ passing through the point $(2, 1)$. [5+5]

4.a) Solve $(2x + 5)^2 \frac{d^2y}{dx^2} - 6(2x + 5) \frac{dy}{dx} + 8y = 6x$.

b) Solve $\frac{d^2y}{dx^2} - y = x \sin x + x^2 e^x$. [5+5]

OR

5.a) Apply the method of variation of parameters to solve $\frac{d^2y}{dx^2} + y = \tan x$.

b) Solve $\frac{d^2y}{dx^2} - 2 \frac{dy}{dx} + 5y = 0, y(0) = -3$ and $\frac{dy}{dx} = 1$ at $x = 0$. [5+5]

6.a) Find the Laplace transform of the Half wave rectifier given as

$$f(t) = \begin{cases} \sin \omega t, & 0 < t < \frac{\pi}{\omega} \\ 0, & \frac{\pi}{\omega} < t < \frac{2\pi}{\omega} \end{cases}$$

b) Solve by using the Laplace transform $y'' + 2y' + 5y = e^{-t} \sin t, y(0) = 0$, and $y'(0) = 1$. Here $y' = \frac{dy}{dt}$. [5+5]

OR

7.a) Evaluate $L^{-1} \left\{ \frac{e^{-s} - 3e^{-3s}}{s^2} \right\}$.

b) Evaluate $\int_0^t e^{-t} \left(\frac{\cos at - \cos bt}{t} \right) dt$ with the help of Laplace transform. [5+5]

8.a) If $\vec{v} = \frac{x\vec{i} + y\vec{j} + z\vec{k}}{\sqrt{x^2 + y^2 + z^2}}$, find the value of divergence ($\vec{v}$).

b) Find the value of n for which the vector $r^n \vec{r}$ is solenoidal, where $\vec{r} = x\vec{i} + y\vec{j} + z\vec{k}$. [5+5]

OR

9.a) Find the divergence and curl of $\vec{v} = (xyz)\vec{i} + (3x^2y)\vec{j} + (xz^2 - y^2z)\vec{k}$ at $(2, -1, 1)$.

b) Prove that $(y^2 - z^2 + 3yz - 2x)\vec{i} + (3xz + 2xy)\vec{j} + (3xy - 2xz + 2z)\vec{k}$ is both solenoidal and irrotational. [5+5]

10.a) A vector field is given by $\vec{F} = (\sin y)\vec{i} + x(1 + \cos y)\vec{j}$. Evaluate the line integral over a circular path $x^2 + y^2 = a^2, z = 0$.

b) Evaluate $\iint (yz\vec{i} + zx\vec{j} + xy\vec{k}) \cdot d\vec{s}$ over S , where S is the surface of the sphere $x^2 + y^2 + z^2 = a^2$ in the first octant. [5+5]

OR

11.a) Use Green's Theorem, evaluate $\int (x^2 y dx + x^2 dy)$ over C , where C is the boundary described counter clockwise of the triangle with vertices $(0, 0), (1, 0), (1, 1)$.

b) Apply Stoke's Theorem to find the value of $\int (y dx + z dy + x dz)$ over C , where C is the curve of intersection of $x^2 + y^2 + z^2 = a^2$ and $x + z = a$. [5+5]

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Max. Marks: 60

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